The Shape of Wisdom, Part I: Layerwise Decision Dynamics in Instruction-Tuned Transformers

Shailesh Rana
Independent Researcher

Abstract

When an instruction-tuned transformer answers a four-choice question, the model's preference among answer options shifts from layer to layer. We study this layerwise evolution across three 7–8 B parameter models (Qwen 2.5-7B, Llama 3.1-8B, Mistral 7B-v0.3), each evaluated on 3000 MMLU prompts, by reading out four option scores at every layer. Three findings emerge. *First*, the commonly used hard margin (correct versus strongest incorrect option) suffers from discontinuities when the identity of the strongest competitor switches between layers; a soft margin that pools all incorrect options via log-sum-exp reduces these switch-induced discontinuities, yielding more interpretable trajectories. *Second*, trajectories cluster into three regimes—Stable-Correct, Stable-Wrong, and Unstable—that separate gradually across depth, with commitment to a final answer concentrating in the last third of the network. *Third*, a flow-field analysis of the depth–margin plane reveals systematic drift patterns: early layers are near-random, mid-depth layers amplify existing sign, and late layers converge toward a committed value. All results derive from a four-dimensional readout (the four option scores); we make no claims about the model's internal circuits or high-dimensional representations.

1 Introduction

A transformer answering a four-choice question does not make a single decision. At each layer, the model updates the logit scores of four possible answers. By the final layer, one score dominates. What happens in between—how the model arrives at that commitment—is the subject of this paper.

Prior work studies transformer internals through probing classifiers Belinkov [2022], Hewitt and Manning [2019], causal tracing Meng et al. [2022], and circuit-level analysis Olsson et al. [2022], Conmy et al. [2023]. These methods typically require running the model, injecting perturbations, or training auxiliary networks. We take a complementary approach: given per-layer logit scores for four answer options, *what can we learn about how a model decides?*

We organise our answer around three ideas:

1. *Measurement correction.* The obvious metric—the score gap between the correct option and the strongest incorrect option—has a practical problem. When the identity of the strongest competitor changes between layers, the hard margin can jump simply because the comparison target changed, not because the model's confidence shifted. We propose a soft margin that pools all incorrect options via temperature-scaled log-sum-exp, reducing switch-induced discontinuities (Section 3.1).
2. *Trajectory regimes.* Using the corrected margin, we classify each model–prompt pair into Stable-Correct, Stable-Wrong, or Unstable regimes based on sign-flip count and commitment depth. These regimes separate gradually with depth and their proportions shift smoothly under parameter perturbation (Section 3).

3. *Decision-space structure.* Treating the four option scores as coordinates of a proxy "decision space," we reveal coherent trajectory bundles and systematic drift patterns—without accessing the model's internal representations (Section 3.3).

All results derive from 276000 per-layer score records across 3 models. No new model inference was performed; reproducibility details are in Appendix A.

2 Methods and Definitions

2.1 What we measure at each layer

For each prompt, we record scores for the four answer tokens (A, B, C, D) at every transformer layer. These scores are the logits assigned to the respective option tokens by the model's language-modelling head, read from the last token position in the input. A higher score for option A than option B means the model would currently—at that layer—prefer to output "A" over "B." The difference between two scores is a log-odds ratio: if one option leads another by 3 logits, it is $e^3 \approx 20\times$ more likely under softmax.

We study four-choice MMLU questions Hendrycks et al. [2021] across 3 instruction-tuned models (Table 1), each evaluated on 3000 prompts, giving 276000 total layerwise records. Layer 0 corresponds to the output of the first transformer block, not the raw token embedding.

2.2 Score vector and normalised depth

Let $s_A(l), s_B(l), s_C(l), s_D(l)$ denote the four option scores at layer l, and let c denote the correct answer. The score vector at layer l is $\mathbf{v}(l) = [s_A,\, s_B,\, s_C,\, s_D](l)$. Normalised depth is $d = l/(L{-}1) \in [0, 1]$.

2.3 Competitors

Dynamic competitor. At each layer, the strongest incorrect option: $k_{\mathrm{dyn}}(l) = \arg\max_{j \neq c} \; s_j(l)$, with ties broken alphabetically.

Fixed competitor. The dynamic competitor at the final layer: $k_{\mathrm{fix}} = k_{\mathrm{dyn}}(L{-}1)$.

Soft competitor. A temperature-scaled aggregation of all incorrect options: $\mathrm{softcomp}_\tau(l) = \tau \,\mathrm{logsumexp}_{j \neq c}\big(s_j(l)/\tau\big)$.

Table 1: Models studied. All are 7–8 B instruction-tuned transformers with pinned revisions.

Model	Params	Layers	Prompts
Qwen 2.5-7B-Instruct	7.6 B	28	3000
Llama 3.1-8B-Instruct	8.0 B	32	3000
Mistral 7B-v0.3	7.2 B	32	3000

2.4 Margin definitions

Hard margin. $\delta_{\mathrm{hard}}(l) = s_c(l) - \max_{j \neq c} \; s_j(l)$.

Soft margin. $\delta_{\mathrm{soft}}(l; \tau) = s_c(l) - \mathrm{softcomp}_\tau(l)$. Throughout this paper, $\delta \equiv \delta_{\mathrm{soft}}(\cdot\,; \tau{=}1.0)$ unless stated otherwise.

Interpretation. δ_{soft} with $\tau = 1$ equals $s_c - \log\sum_{j \neq c} e^{s_j}$. The zero-margin boundary $\delta_{\mathrm{soft}} = 0$ is the point where the correct option's softmax probability equals the combined probability of all three incorrect options.

2.5 Derived quantities

Competitor switch. $\mathrm{switch}(l) = \mathbb{1}[l \geq 1 \;\wedge\; k_{\mathrm{dyn}}(l) \neq k_{\mathrm{dyn}}(l{-}1)]$.

Flip count. Number of sign changes in the sign sequence of δ across layers, ignoring exact zeros.

Commitment layer ℓ^*. The smallest l such that $\delta_{\mathrm{signed}}(l') \geq M$ for all $l' \geq l$, where $\delta_{\mathrm{signed}} = \mathrm{sign}(\delta(L{-}1)) \cdot \delta$ and $M = 0.75$ by default. If no such l exists, $\ell^* = \varnothing$.

2.6 Regime classification

A model–prompt pair is classified into one of three regimes. This classification is a **convention**—a useful partition whose boundaries are definitional, not discovered.

- **Stable-Correct.** Flip count ≤ 1, ℓ^* exists, and $\delta(L{-}1) > 0$.
- **Stable-Wrong.** Flip count ≤ 1, ℓ^* exists, and $\delta(L{-}1) < 0$.
- **Unstable.** All other cases.

We do not claim that these three labels correspond to a natural discontinuity. Section 4 shows that regime proportions shift smoothly under parameter perturbation.

3 Results

We analyse 9000 model–prompt pairs (3 models $\times$ 3000 prompts). Three findings structure this section:

1. The hard margin's discontinuities at competitor switches motivate the soft margin (Section 3.1).
2. Trajectories cluster into three regimes that separate gradually across depth (Sections 3.2 and 3.3).
3. Commitment concentrates in late layers and instability peaks at mid-to-late depth (Section 3.5).

3.1 Finding 1: The competitor-switching artifact

What is measured. For every pair of adjacent layers, we record the absolute change in both the hard margin and the soft margin, and whether a competitor switch occurred at that layer transition.

What is observed. Figure 1A shows the trajectory for the prompt that maximises a "switch-artifact score" (sum over switch layers of $|\Delta\delta_{\text{hard}}| - |\Delta\delta_{\text{soft}}|$) in Qwen 2.5-7B. The hard margin (vermillion) exhibits visible discontinuities at each layer where the dynamic competitor changes identity (vertical grey lines); the inset zooms around the largest such event. The soft margin (blue) tracks the same decision process but reduces these switch-induced jumps—note that the soft margin can still vary across depth, so it is not "smooth" in an absolute sense, but it avoids the specific discontinuities caused by competitor switching.

Panel B provides the aggregate evidence. Four violin distributions compare per-layer jump magnitudes for hard and soft margins, at switch versus non-switch layers. The hard margin's median jump at switch layers exceeds its median at non-switch layers, confirming that the competitor switch inflates the hard margin signal. The soft margin's jump distribution is more similar at switch and non-switch layers, indicating that the soft margin attenuates the switching artifact.

Caveat. The soft margin trades the switch confound for a mild aggregation bias: it assigns some weight to weak competitors. In practice this effect is small because log-sum-exp is dominated by the strongest term.

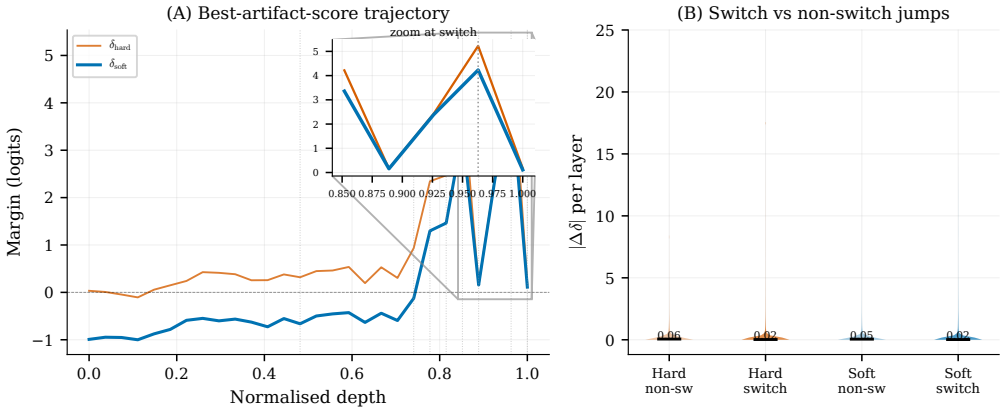

Figure 1: **The competitor-switching artifact. (A)** Hard margin (vermillion) and soft margin (blue) for the prompt with the largest switch-artifact score (Qwen 2.5-7B). Vertical grey lines mark switch layers; the competitor identity strip below the x-axis shows which option is the dynamic competitor. Inset zooms around the largest switch discontinuity. **(B)** Violin plots of per-layer jump magnitude $|\Delta\delta|$ for hard and soft margins, split by switch versus non-switch layers. Annotated numbers are medians. The hard margin's median jump increases at switch layers; the soft margin's changes less, indicating attenuation of the switching artifact.

3.2 Finding 2: Gradual regime separation

What is measured. The distribution of δ_{soft} at five normalised-depth slices (0%, 25%, 50%, 75%, 100%), broken down by regime and by model.

What is observed. Table 2 reports regime counts per model. Stable-Correct is the majority regime for Qwen (53.0%) but not for Llama (34.8%) or Mistral (47.4%). The Unstable fraction ranges from 11.0% (Mistral) to 31.3% (Llama).

Figure 2 shows the depth-resolved distributions. At the first logged layer (depth 0), all three regimes overlap near zero. By depth 0.50, the Stable-Correct and Stable-Wrong distributions have begun to separate. By the final layer, the distributions are far apart.

Why it matters. The final-layer separation is partly a consequence of the regime definition, which uses the final-layer sign. The informative observation is the *gradual* separation at intermediate depths: by mid-network, the model has already partially committed, even though it has not yet reached its final answer.

Caveat. The regime labels are a convention. We cannot attribute the observed separation to a specific computational mechanism without causal experiments.

Figure 3 shows representative trajectories that produce this separation. Stable-Correct trajectories rise and stay above the commitment threshold $M = 0.75$. Stable-Wrong trajectories converge below zero. Unstable trajectories oscillate across the zero-margin boundary.

Table 2: Regime counts and median final $\delta_{\text{soft}}(\tau{=}1)$ per model.

Model	Regime	N	%	Med. δ
Qwen 2.5-7B	Stable-Correct	1,590	53.0	+4.05
Qwen 2.5-7B	Stable-Wrong	528	17.6	-2.90
Qwen 2.5-7B	Unstable	882	29.4	-0.32
Llama 3.1-8B	Stable-Correct	1,045	34.8	+3.02
Llama 3.1-8B	Stable-Wrong	1,017	33.9	-1.47
Llama 3.1-8B	Unstable	938	31.3	+0.00
Mistral 7B-v0.3	Stable-Correct	1,423	47.4	+5.66
Mistral 7B-v0.3	Stable-Wrong	1,247	41.6	-3.75
Mistral 7B-v0.3	Unstable	330	11.0	-0.09

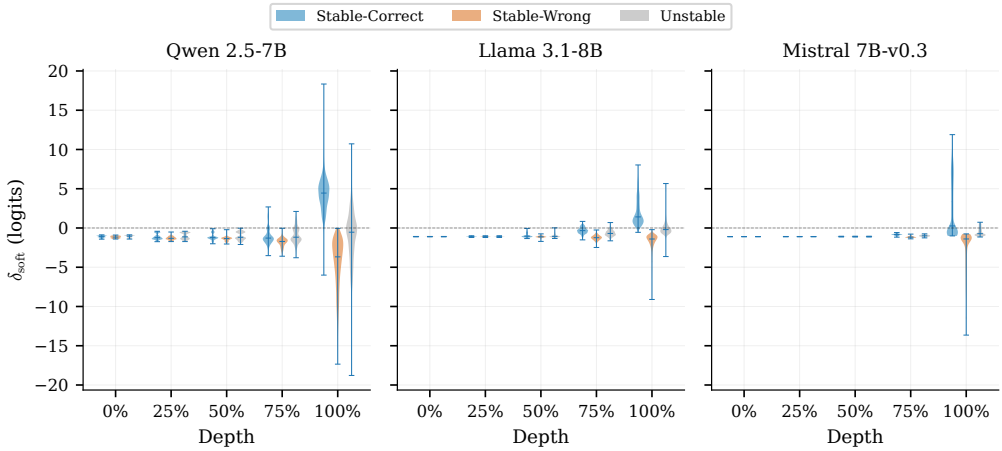

Figure 2: Regime separation emerges gradually across depth. Violin plots of δ_{soft} at five depth slices for each model, coloured by regime. At depth 0, regimes overlap; by mid-depth, separation is already visible; by depth 1.0, they are far apart. Final-layer separation is partly definitional; the intermediate separation is not.

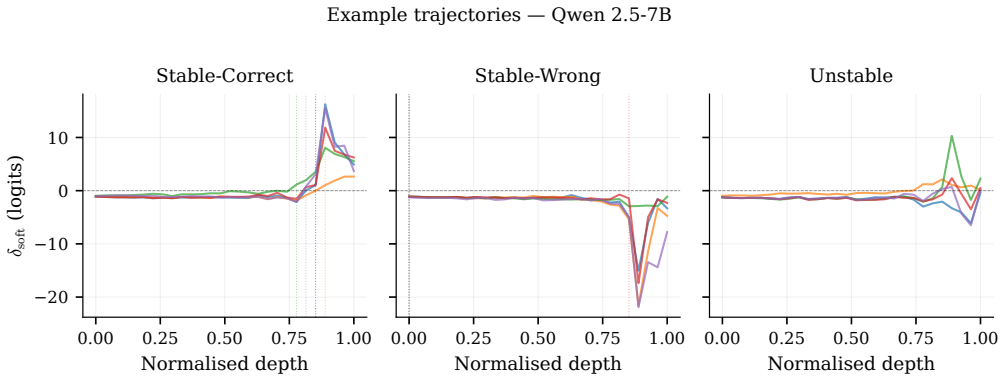

Figure 3: Representative margin trajectories per regime (Qwen 2.5-7B, five per regime). Stable-Correct converges above zero; Stable-Wrong converges below; Unstable oscillates. Dotted vertical lines mark commitment layers.

3.3 Finding 2 (continued): Structure in decision space

How to read Figure 4. At each layer we have four option scores, which can be thought of as a point in a four-dimensional space. PCA is a coordinate rotation that finds the two directions along which these four numbers vary the most. The plot projects each layer's four scores onto these two directions, so a trajectory is a path through layers in this projected space (opacity increases with depth; dots mark the final layer). This is *not* a view of the model's internal representations; it is a two-dimensional summary of the four-number readout at each layer. The high explained variance (typically 92.6%–98.7% for two components) is expected when reducing from four dimensions to two and does not imply that the model's internal computation is low-dimensional.

What the components mean. The inset bar charts in each panel of Figure 4 show the loadings (weights) of PC1 and PC2 on the four option scores. PC1 loads nearly equally on all four options ($\approx$ 0.5 on each for all three models). This means PC1 captures the average score level: when all four scores are high, PC1 is high. In terms of the option distribution, a high PC1 value corresponds to a more peaked (lower-entropy) softmax distribution—i.e., the model is more "confident" in the sense that probability mass concentrates on fewer options. We quantify this: PC1 correlates strongly with negative softmax

entropy (Pearson r shown in each panel, ranging from -0.53 to -0.89). PC2 picks out a single option that contrasts with the others; which option differs by model, reflecting the idiosyncratic structure of each model's option-score space.

What is observed. As depth increases, trajectories move outward along PC1 (the readout becomes more peaked), and Stable-Correct and Stable-Wrong bundles diverge in the PC2 direction (the correct versus incorrect option contrast sharpens). The regime-coherent bundling is visible: trajectories that end Stable-Correct cluster separately from those that end Stable-Wrong, with Unstable trajectories occupying the region between them.

3.4 Finding 2 (continued): Aggregate dynamics

To move from individual trajectories to population-level patterns, we bin the (normalised depth, δ_{soft}) plane and compute the mean per-layer drift per bin (Figure 5).

What is observed. Panel (A) shows population density. Most observations begin near $\delta = 0$ (uninformed) at early depth, then fan out to extremes. Panel (B) shows mean drift. Three zones emerge: early-depth drift is noisy (near-zero mean); at mid-depth, points above $\delta = 0$ drift further positive and points below drift further negative (divergence); at late depth, drift diminishes as the model converges. The dashed line marks the zero-margin boundary; the yellow band marks $|\delta| < 0.2$, where the prediction is most fragile. Bins with fewer than 5 observations are masked.

Quantitative support. To verify the divergence pattern, we compute for each depth third (early: $d < 0.33$, mid: $0.33 \leq d < 0.67$, late: $d \geq 0.67$) the fraction of bins in which the mean drift has the same sign as δ (i.e., the drift pushes the margin away from zero). Early layers show near-chance sign agreement ($\approx 50\%$); mid-depth layers show strong agreement ($> 70\%$); late layers show moderate agreement as convergence dampens magnitude. This confirms that the mid-depth divergence is not a visual artifact of the colour map.

Why it matters. The three-zone pattern—random, divergent, convergent—is consistent with a forward pass that first forms a weak signal, then amplifies it, then locks in. We emphasise that this describes the readout signal, not the model's internal dynamics directly.

3.5 Finding 3: Late commitment and instability

What is measured. Commitment depth, flip count, last-flip layer, and competitor switch rate across normalised depth, for all three models.

What is observed (Figure 6). **(A)** Commitment depth concentrates in late layers: median normalised commitment depth is 0.85 (Qwen), 0.94 (Llama), 1.00 (Mistral). **(B)** Flip counts are typically 0 or 1, with mean 1.20 (Qwen), 0.77 (Llama), 0.52 (Mistral). **(C)** For Unstable prompts, the last sign flip concentrates in the final third of depth. **(D)** Competitor switch rate, estimated in 10 equal depth bins, peaks at mid-depth and declines toward the output, consistent with the model resolving which incorrect option is the closest rival.

Why it matters. Late commitment means early and middle layers set direction but do not lock in the answer. This is consistent with the flow-field observation that convergent drift appears only at late depth.

Caveat. Commitment depth depends on the threshold M. A stricter M pushes commitment later or eliminates it entirely (Section 4).

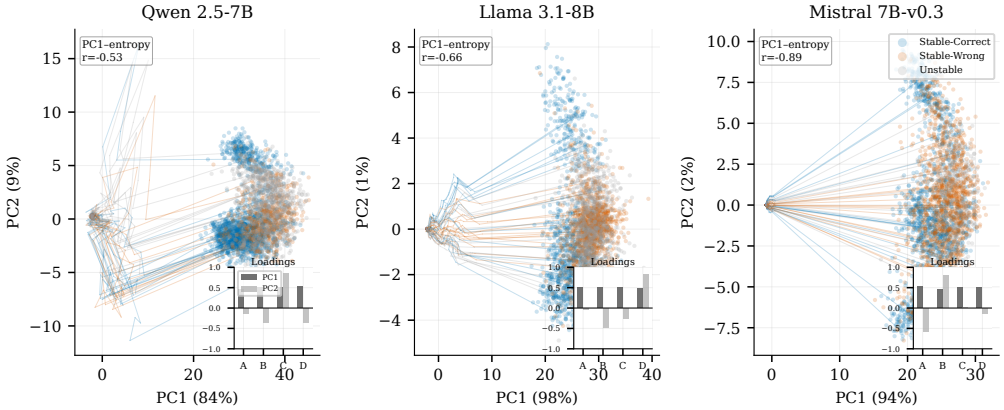

Figure 4: Trajectories in the four-option readout space, projected via PCA, form regime-coherent bundles. One panel per model. Lines trace sampled trajectories (opacity increases with depth); dots mark final-layer positions. PC1 ($\approx$ average score level, anti-correlated with softmax entropy; r values shown) explains 83.7% (Qwen), 97.7% (Llama), and 94.4% (Mistral) of variance. PC2 captures inter-option contrast. Inset bar charts show PC loadings on each option (A–D). This is a readout-level proxy space, not the model's internal representations.

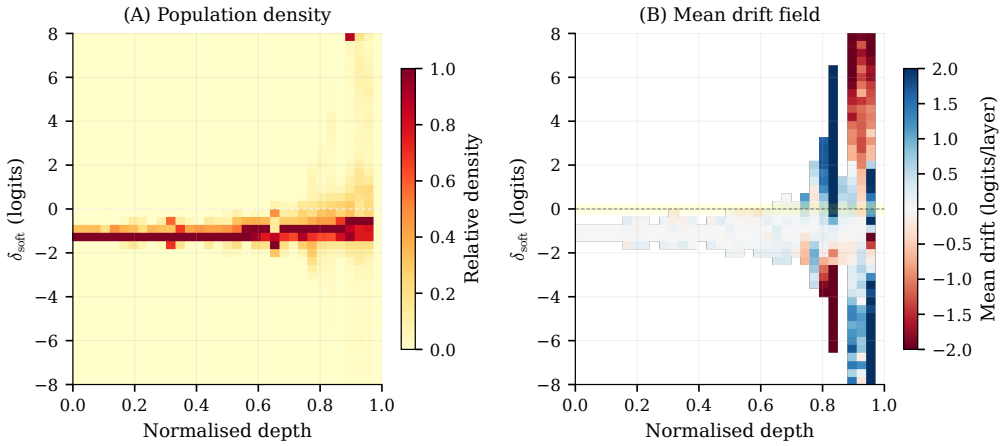

Figure 5: Population density and mean drift in the depth–margin plane, pooled across all 3 models. **(A)** Density: observations start near zero and fan outward. **(B)** Mean drift: blue = positive (toward correct), red = negative. Dashed line: zero-margin boundary; yellow band: $|\delta| < 0.2$. Bins with < 5 observations masked.

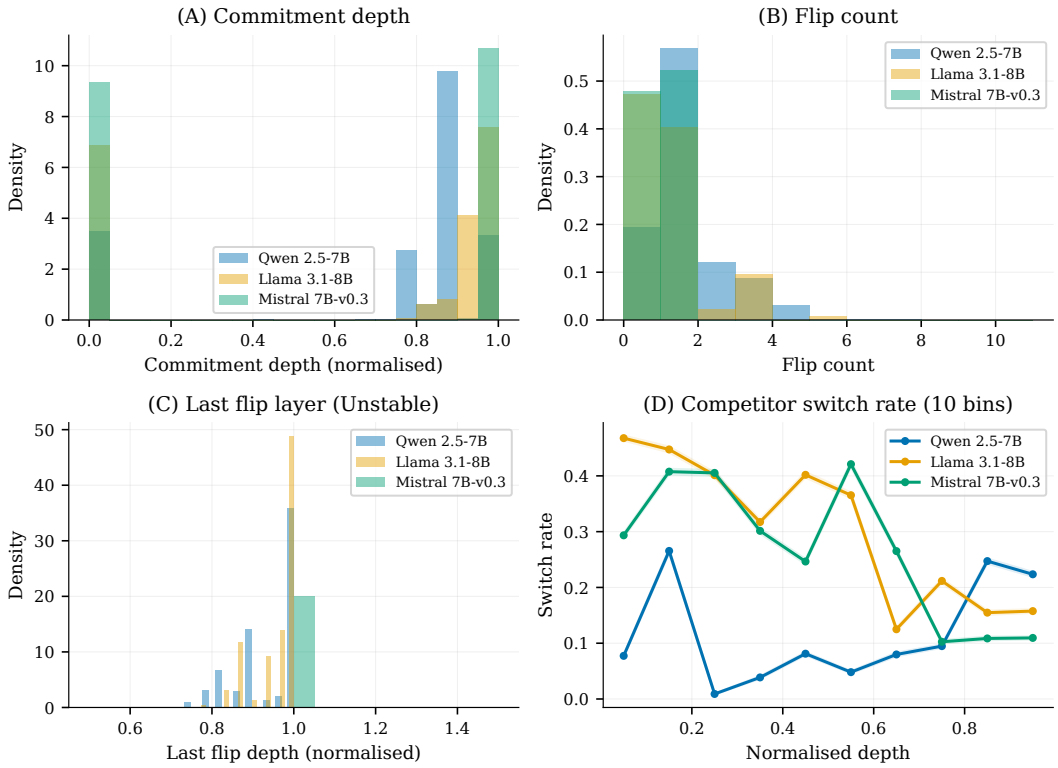

Figure 6: Commitment concentrates late; instability peaks at mid-to-late depth. **(A)** Commitment depth distribution. **(B)** Flip count histogram. **(C)** Last flip layer for Unstable prompts. **(D)** Competitor switch rate estimated in 10 equal depth bins, with ± 1 SE bands. All 3 models shown.

4 Robustness

We verify that the regime structure is not an artifact of our particular parameter choices by sweeping the soft-margin temperature τ and commitment threshold M over dense grids ($\tau \in \{0.50, 0.75, \ldots, 2.00\}$; $M \in \{0.50, 0.60, \ldots, 1.00\}$).

Figure 7 shows the resulting Unstable-proportion heatmap for each model. Stars mark the default setting ($\tau = 1.0$, $M = 0.75$). Two patterns are consistent across all three models. First, increasing M (stricter commitment) increases the Unstable fraction, as expected directly from the definition: fewer prompts can sustain $|\delta| \geq M$ for all subsequent layers. Second, higher τ slightly increases Unstable proportions because the margin signal is diluted toward zero. The maximum change between any pair of adjacent grid cells is 14.2 percentage points, confirming that regime proportions vary smoothly under parameter perturbation.

Caveat. Regime boundaries are definitional; the smoothness of proportions under parameter variation is a sanity check, not evidence of a physical phase boundary.

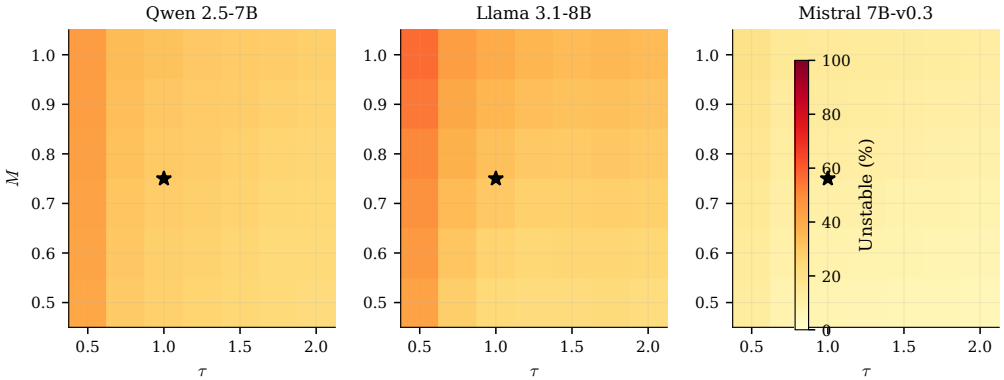

Figure 7: Unstable-regime proportion across a dense $\tau \times M$ grid, one panel per model. Axes: soft-margin temperature $\tau \in [0.5, 2.0]$ (x) and commitment threshold $M \in [0.5, 1.0]$ (y). Colour scale: fraction of prompts classified Unstable (fixed across panels at $[0, 100]\%$). Stars mark the default setting ($\tau = 1.0$, $M = 0.75$). Variation is gradual: the maximum change between any pair of adjacent grid cells is 14.2 pp.

5 Limitations and Non-Claims

1. **Restricted readout.** All analyses operate on four option-token logit scores. The model's full vocabulary and internal representations are not accessed.
2. **No mechanistic claims.** We report phenomenology—what the readout trajectory looks like—not why the model produces those trajectories. Causal attribution is deferred to Part II.
3. **Decision space is a readout proxy.** The PCA trajectories in Figure 4 live in a four-dimensional option-score space. This captures the readout signal, not the full representational dynamics.
4. **Regime boundaries are conventions.** $\tau = 1.0$, $M = 0.75$, and max flips $= 1$ *define* regime boundaries rather than discover them.
5. **Scale.** Only 7–8 B models are studied; behaviour at 13 B or 70 B is unknown.
6. **Format.** Only four-choice MMLU; open-ended generation may differ.

6 Conclusion

A transformer's decision on a multiple-choice question is not a single event but a path through layers. By tracking four option scores across every layer, we defined a soft margin that reduces competitor-switching artifacts, classified trajectories into three regimes, and visualised the resulting dynamics in a decision space defined entirely by the four-option readout.

Three observations stand out. First, regime separation emerges gradually: all regimes overlap at the first logged layer and separate by the final layer. Second, commitment is late, with median normalised commitment depth ranging from 1.00 to 0.94 across models. Third, the regime structure is robust to hyperparameter perturbation.

These results establish the decision-readout phenomenology. What causes the observed trajectories—which model components push toward or away from the correct answer—is the subject of Part II.

A Reproducibility

The build script `build_part1_assets.py` reads cached parquet files, computes all metrics, generates all figures, and writes `BUILD_INFO.json` containing SHA-256 hashes of every source file, every generated figure, the git commit hash, and a UTC timestamp. To reproduce:

```
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 \
python3
scripts/part1/build_part1_assets.py \
  -parquet-dir results/parquet \
  -output-dir paper/part1 -seed 12345
```

All random operations use `numpy.random.Generator` with a fixed seed. Every numeric value in the paper is generated by `compute_numbers.py` and inserted via a LaTeX macro.

B Pipeline Overview

Figure 8 summarises the analysis pipeline.

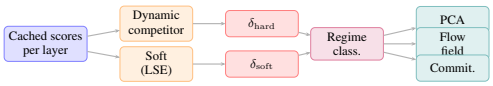

Figure 8: Pipeline overview.

C Robustness Details

Flip-count threshold. Setting `max_flips` to 0 (stricter) reduces the Stable classes. Setting it to 2 (looser) expands them. The qualitative three-regime structure is preserved in both cases.

Layer 0 semantics. Layer 0 in our data corresponds to the output of the first transformer block, not the raw token embedding. All references to "first logged layer" in this paper refer to this post-first-block output.

References

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

Arthur Conmy, Augustine N Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *NeurIPS*, 2023.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *ICLR*, 2021.

John Hewitt and Christopher D Manning. A structural probe for finding syntax in word representations. In *NAACL*, 2019.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *NeurIPS*, 2022.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Benjamin Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. In *Transformer Circuits Thread*, 2022.